

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I **G. Ramya/192572073** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *G. Ramya*

Assessment Tasks

Class Test

1. Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.
2. Explain the 5Vs of Big Data with one practical example for any two Vs.
3. Differentiate structured, semi-structured, and unstructured data with examples.
4. Explain the role of distributed file systems in big data processing.
5. List any three major security concerns in big data platforms and briefly explain each.
6. What are data appliances and integration tools? Explain their role in big data workflows.
7. Discuss two major challenges organizations face while implementing big data analytics.
8. Write short notes on any one domain application of big data analytics.
9. How do business drivers influence the adoption of big data solutions?
10. Explain the importance of big data analytics in decision making.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful	Some relevant examples	Weak examples	No useful examples

		examples			
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1. Define Big Data and explain why traditional data processing approaches are often insufficient.

Big Data refers to extremely large, complex, and rapidly generated datasets that cannot be efficiently stored, processed, or analyzed using traditional data-processing systems.

Traditional systems are often insufficient because:

- They have limited storage capacity.
- They cannot process very high data volumes quickly.
- They mainly handle structured data.
- Scaling them can be expensive and difficult.
- They may not support real-time processing of continuously generated data.

Example: Social media platforms generate millions of posts, images, videos, and interactions every day, requiring distributed big data technologies.

2. Explain the 5Vs of Big Data with examples.

The **5Vs** are:

1. **Volume** – Refers to the huge amount of data generated.
 - *Example:* An e-commerce website stores millions of customer transactions.
2. **Velocity** – Refers to the speed at which data is generated and processed.
 - *Example:* Banking systems process thousands of transactions every second.
3. **Variety** – Refers to different types and formats of data.
 - Examples: Text, images, videos, JSON, and database records.
4. **Veracity** – Refers to the accuracy and reliability of data.
 - Incorrect or duplicate customer records reduce data veracity.
5. **Value** – Refers to the useful information obtained from data.
 - *Example:* Analyzing customer purchases can help an online store recommend suitable products.

3. Differentiate structured, semi-structured, and unstructured data.

Type	Meaning	Examples
Structured	Organized in a fixed format such as rows and columns	SQL tables, employee records
Semi-structured	Does not follow a strict table structure but has tags or keys	JSON, XML, emails
Unstructured	Has no predefined structure	Images, videos, audio, social-media posts

Example: A hospital may store patient details in a database, insurance claims in JSON, and X-ray images as unstructured data.

4. Explain the role of distributed file systems in big data processing.

A **Distributed File System (DFS)** stores large files by dividing them into smaller blocks and distributing those blocks across multiple computers.

Roles:

- Stores very large datasets.
- Provides scalability by adding more machines.
- Supports parallel data processing.
- Provides fault tolerance through data replication.
- Improves data availability.

Example: HDFS (Hadoop Distributed File System) is commonly used for storing and processing large datasets in Hadoop environments.

5. List any three major security concerns in big data platforms.

1. Data Privacy:

Sensitive information such as personal, financial, or medical data must be protected from unauthorized access.

2. Unauthorized Access:

Attackers or unauthorized users may access databases or data lakes. Authentication and access control help prevent this.

3. Data Breaches:

Large centralized datasets are attractive targets for attackers. Encryption, monitoring, and security controls help reduce the risk.

6. What are data appliances and integration tools?

Data appliances are specialized hardware or software systems designed to store, manage, and process large amounts of data efficiently.

Integration tools connect data from different sources and move or transform it into a common system.

Roles in big data workflows:

- Collect data from multiple sources.
- Extract and transform data.
- Clean and combine data.
- Transfer data to storage systems.
- Support real-time and batch data processing.

Examples: Apache Kafka for data streaming and ETL tools for data integration.

7. Discuss two major challenges organizations face while implementing big data analytics.

1. Data Security and Privacy:

Organizations handle sensitive information and must protect it from unauthorized access, breaches, and misuse.

2. Lack of Skilled Professionals:

Big data requires knowledge of distributed systems, databases, cloud platforms, analytics, and machine learning. Finding skilled professionals can be difficult.

Other challenges include high implementation costs, data quality problems, and integrating different data sources.

8. Short note on one domain application of big data analytics.

Healthcare

Big data analytics is widely used in healthcare.

Applications:

- Analyzing patient records.
- Predicting disease risks.
- Supporting medical research.
- Monitoring patients.
- Analyzing medical images.
- Improving hospital resource management.

Example: Historical patient data can be analyzed to identify patterns that help predict the risk of certain diseases.

9. How do business drivers influence the adoption of big data solutions?

Business drivers are the factors that encourage an organization to adopt big data technologies.

Major drivers include:

- **Better decision making:** Data provides useful business insights.
- **Customer understanding:** Companies can analyze customer behavior.
- **Cost reduction:** Analytics can identify inefficient processes.
- **Revenue growth:** Data can reveal new business opportunities.
- **Competitive advantage:** Organizations can respond faster to market changes.
- **Operational improvement:** Real-time analytics can improve business processes.

Thus, organizations adopt big data when it can provide measurable business value.

10. Explain the importance of big data analytics in decision making.

Big Data Analytics helps organizations examine large datasets and discover useful patterns, trends, and relationships.

Importance:

- Provides data-driven decisions.
- Helps predict future trends.
- Identifies customer preferences.
- Detects fraud and unusual activities.
- Improves operational efficiency.
- Reduces business risks.
- Helps organizations develop better products and services.

Example: A retail company can analyze previous purchases to predict customer demand and maintain appropriate stock levels.

Conclusion: Big data analytics converts large amounts of raw data into useful information, helping organizations make **faster, smarter, and more informed decisions**.